

ENGINEERING DESIGN & INNOVATION

Project Theme Leave and Payroll Management System

1. Introduction

1.1 Project Overview

The Leave and Payroll Management System (LPMS) is a mini project that automates two closely linked HR functions — employee leave administration and monthly salary processing — for a small-to-medium organization. The system allows employees to apply for leave, managers to approve requests, and HR/Admin to configure salary components and generate payslips through a rule-driven calculation engine.

1.2 Problem Statement

Manual handling of leave records and salary computation is error-prone, time-consuming and difficult to audit. Leave balances are often tracked in spreadsheets disconnected from payroll, causing incorrect Loss-of-Pay deductions and delayed payslips. This project designs and implements an integrated system that stores employee, leave and payroll data in a normalized database, applies configurable business rules to compute pay, and produces standard HR reports.

1.3 Scope

- Employee master data management (departments, designations).
- Leave application, approval workflow and automatic balance tracking.
- Attendance capture feeding Loss-of-Pay into payroll.
- Configurable rule-based payroll calculation (earnings, deductions, tax slabs).
- Report generation: payslip, payroll register, leave and attendance summaries.
- Role-based access for Employee, Manager and HR/Admin.

2. Module Breakdown

1. Employee Management Module

Maintains master data for employees, departments and designations; the foundation entity referenced by every other module.

Key Functions

- Add / update / deactivate employee records
- Assign department, designation and reporting manager
- Maintain user accounts and roles

Foundational Subjects Applied

OOP (Employee class hierarchy), DBMS (Employee/Department normalized tables), Software Engineering (CRUD use cases).

2. Authentication & Authorization Module

Controls who can log in and what each role (Employee / Manager / HR-Admin) can see or do.

Key Functions

- Login / session or token management
- Role-based access control
- Password hashing & validation

Foundational Subjects Applied

Web Technology (sessions/JWT, REST), OOP (Role/User classes, interfaces).

3. Leave Management Module

Handles leave application, the manager approval workflow, and automatic leave-balance updates.

Key Functions

- Apply for leave
- Approve / reject via workflow queue
- Auto-update LEAVE_BALANCE on approval
- View leave history

Foundational Subjects Applied

Data Structures (approval queue, balance hash map), DBMS (triggers for balance update), Software Engineering (workflow use case/DFD).

4. Attendance Module

Captures daily attendance, feeding Loss-of-Pay (LOP) inputs to the payroll engine.

Key Functions

- Mark / import daily attendance
- Compute monthly present/absent/LOP summary

Foundational Subjects Applied

DBMS (attendance table joins & aggregation), Data Structures (arrays/lists for daily records).

5. Payroll & Rule Engine Module (core)

Computes gross pay, deductions and net pay by applying configurable calculation rules (tax slabs, PF/ESI, LOP, overtime).

Key Functions

- Configure salary components & rules
- Evaluate rule set per employee (Strategy pattern)
- Run monthly payroll generating payslips

- Handle pro-rata / arrear recalculation

Foundational Subjects Applied

PPL (rule/decision-table evaluation, expression evaluation), OOP (Strategy pattern for pluggable rules), DBMS (transactional payroll run).

6. Report Generation Module

Produces the printable/exportable reports consumed by employees, managers and HR/finance.

Key Functions

- Generate individual Payslip
- Generate monthly Payroll Register
- Generate Leave Summary & Attendance reports

Foundational Subjects Applied

DBMS (aggregate queries, grouping), Software Engineering (report/UI design), Web Technology (rendering/exporting reports).

7. Admin / Configuration Module

Lets HR/Admin configure master data that the rule engine and other modules depend on.

Key Functions

- Define leave types & entitlements
- Define/modify salary components and tax slab rules
- Manage departments & roles

Foundational Subjects Applied

DBMS (master-data schema), OOP (configuration objects), Software Engineering (change/version control of business rules).